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package comprog_project;

import javax.swing.*.*;
import javax.swing.border.EmptyBorder;
import java.awt.*.*;
import java.awt.event.*;
import java.text.SimpleDateFormat;
import java.util.ArrayList;
import java.util.Date;
import java.util.List;
import javax.swing.table.DefaultTableModel;
import javax.swing.table.JTableHeader;
import javax.swing.table.TableRowSorter;

public class Reports extends JFrame {

    private JPanel contentPane;
    private JTextField searchField;
    private JTable ordersTable;
    private DefaultTableModel tableModel;
    private JComboBox<String> comboBoxSales;
    private TableRowSorter<DefaultTableModel> sorter;
    private List<String> reportHistory = new ArrayList<>();
    private List<Object[]> originalData = new ArrayList<>();
    private JButton btnGenerateReport, btnViewPastReports;

    public Reports() {
        setTitle("Reports");
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setBounds(100, 100, 900, 600);
        contentPane = new JPanel();
        contentPane.setBackground(Color.WHITE);
        contentPane.setBorder(new EmptyBorder(15, 15, 15, 15));
        setContentPane(contentPane);
        contentPane.setLayout(null);

        JLabel lblReports = new JLabel("Reports");
        lblReports.setFont(new Font("Arial", Font.BOLD, 18));
        lblReports.setBounds(15, 15, 200, 25);
        contentPane.add(lblReports);

        JPanel searchPanel = new JPanel(new BorderLayout());
        searchPanel.setBounds(15, 100, 325, 30);
        searchPanel.setBorder(BorderFactory.createLineBorder(new Color(200, 200, 200)));

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searchField = new JTextField();
searchField.setBorder(BorderFactory.createEmptyBorder(5, 10, 5, 5));
searchField.setToolTipText("Search order ID");

JButton searchButton = new JButton("🔍");
searchButton.setBorder(null);
searchButton.setBackground(Color.WHITE);
searchButton.setFocusPainted(false);

searchPanel.add(searchField, BorderLayout.CENTER);
searchPanel.add(searchButton, BorderLayout.EAST);
contentPane.add(searchPanel);

comboBoxSales = new JComboBox<>();
comboBoxSales.setModel(new DefaultComboBoxModel<>(new String[] {"All Sales",
"KangKong Chips"}));
comboBoxSales.setBounds(400, 100, 150, 30);
contentPane.add(comboBoxSales);

JButton btnFilter = new JButton("Filter");
btnFilter.setBounds(670, 100, 80, 30);
btnFilter.setBackground(new Color(110, 0, 220));
btnFilter.setForeground(Color.WHITE);
btnFilter.setBorder(null);
contentPane.add(btnFilter);

JButton btnClearFilter = new JButton("Clear");
btnClearFilter.setBounds(760, 100, 80, 30);
btnClearFilter.setBackground(Color.WHITE);
btnClearFilter.setBorder(BorderFactory.createLineBorder(new Color(110, 0, 220)));
btnClearFilter.setForeground(new Color(110, 0, 220));
contentPane.add(btnClearFilter);

btnGenerateReport = new JButton("Generate Report");
btnGenerateReport.setBounds(400, 50, 150, 30);
btnGenerateReport.setBackground(new Color(110, 0, 220));
btnGenerateReport.setForeground(Color.WHITE);
contentPane.add(btnGenerateReport);

btnViewPastReports = new JButton("View Past Reports");
btnViewPastReports.setBounds(560, 50, 150, 30);
btnViewPastReports.setBackground(new Color(110, 0, 220));
btnViewPastReports.setForeground(Color.WHITE);

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contentPane.add(btnViewPastReports);

String[] columnNames = {"Select", "Order ID", "Date", "Requested By", "Sales Channel",
"Item", "Items"};
tableModel = new DefaultTableModel(null, columnNames) {
    @Override
    public Class<?> getColumnClass(int column) {
        if (column == 0) return Boolean.class;
        return String.class;
    }

    @Override
    public boolean isCellEditable(int row, int column) {
        return column == 0;
    }
};

addSampleData();

ordersTable = new JTable(tableModel);
ordersTable.setRowHeight(40);
ordersTable.setShowGrid(false);
ordersTable.setSelectionMode(ListSelectionModel.SINGLE_SELECTION);
ordersTable.setBackground(Color.WHITE);

sorter = new TableRowSorter<>(tableModel);
ordersTable.setRowSorter(sorter);

JTableHeader header = ordersTable.getTableHeader();
header.setBackground(Color.WHITE);
header.setFont(new Font("Arial", Font.BOLD, 12));

JScrollPane scrollPane = new JScrollPane(ordersTable);
scrollPane.setBounds(15, 145, 855, 400);
scrollPane.setBorder(BorderFactory.createLineBorder(new Color(220, 220, 220)));
contentPane.add(scrollPane);

btnGenerateReport.addActionListener(e -> generateReport());
btnViewPastReports.addActionListener(e -> viewPastReports());
btnFilter.addActionListener(e -> applyFilter());
btnClearFilter.addActionListener(e -> clearFilters());
searchButton.addActionListener(e -> searchReports());
}

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private void addSampleData() {
    String today = new SimpleDateFormat("MM/dd/yyyy").format(new Date());
    Object[] row1 = {false, "#0001", today, "Josh Mojica", "KangKong Chips", "Cheese", "3"};
    Object[] row2 = {false, "#0002", today, "Josh Mojica", "KangKong Chips", "Sour and
Cream", "3"};
    Object[] row3 = {false, "#0003", today, "Josh Mojica", "KangKong Chips", "Barbeque", "3"};
    Object[] row4 = {false, "#0004", today, "Josh Mojica", "KangKong Chips", "Chocolate", "3"};
    Object[] row5 = {false, "#0005", today, "Josh Mojica", "KangKong Chips", "Spicy Cheese",
"3"};

    originalData.add(row1);
    originalData.add(row2);
    originalData.add(row3);
    originalData.add(row4);
    originalData.add(row5);

    tableModel.addRow(row1);
    tableModel.addRow(row2);
    tableModel.addRow(row3);
    tableModel.addRow(row4);
    tableModel.addRow(row5);
}

private void generateReport() {
    StringBuilder report = new StringBuilder();
    int rowCount = tableModel.getRowCount();
    for (int i = 0; i < rowCount; i++) {
        Boolean isSelected = (Boolean) tableModel.getValueAt(i, 0);
        if (isSelected != null && isSelected) {
            for (int j = 1; j < tableModel.getColumnCount(); j++) {
                report.append(tableModel.getValueAt(i, j)).append(" | ");
            }
            report.append("\n");
        }
    }

    if (report.length() == 0) {
        JOptionPane.showMessageDialog(this, "No rows selected for the report.");
        return;
    }

    String generated = "Report generated on " +
        new SimpleDateFormat("MM/dd/yyyy HH:mm:ss").format(new Date()) +
        "\n" + report.toString();
}

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        reportHistory.add(generated);
        JOptionPane.showMessageDialog(this, "Report generated:\n\n" + generated);
    }

    private void viewPastReports() {
        String historyText = String.join("\n\n", reportHistory);
        JOptionPane.showMessageDialog(this, historyText.isEmpty() ? "No past reports found." :
historyText, "Past Reports", JOptionPane.INFORMATION_MESSAGE);
    }

    private void applyFilter() {
        String filterText = searchField.getText();
        if (!filterText.isEmpty()) {
            sorter.setRowFilter(RowFilter.regexFilter("(?i)" + filterText));
        } else {
            sorter.setRowFilter(null);
        }
    }

    private void clearFilters() {
        searchField.setText("");
        sorter.setRowFilter(null);
    }

    private void searchReports() {
        String searchText = searchField.getText().trim().toLowerCase();
        if (!searchText.isEmpty()) {
            sorter.setRowFilter(RowFilter.regexFilter("(?i)" + searchText));
        }
    }

    public static void main(String[] args) {
        EventQueue.invokeLater(() -> {
            try {
                Reports frame = new Reports();
                frame.setVisible(true);
            } catch (Exception e) {
                e.printStackTrace();
            }
        });
    }
}

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